Comparison of Poststack Seismic Inversion Methods SI1.1

Brian Russell and Dan Hampson, Hampson-Russell Software Services Ltd., Canada*

SUMMARY:

Single-trace post-stack seismic inversion is a processing technique whose aim is to extract acoustic impedance information from stacked seismic data. In principle, the process is straightforward, and involves the assumptions that the seismic trace can be modelled using the convolutional and vertical incidence reflection coefficient equations. In practice, there are many separate methods used to perform post-stack inversion. These methods can be conveniently grouped into three basic categories: 1) classical Recursive methods; 2) sparse-spike methods; and 3) model-based methods.

The three types of inversion will be compared and contrasted using two model datasets, and two real seismic datasets. The models will include a detailed synthetic in which the result is know, as well as the classic wedge model. The real datasets include a gas sand example and a carbonate reef example, both from Alberta. Based on these examples, a number of observations will be made about the various inversion methods.

INTRODUCTION:

Inversion of post-stack seismic data to produce an estimate of the earth's acoustic impedance has been an objective of geophysicists for a number of years, and the published literature contains descriptions of numerous approaches (Lindseth, 1979; Oldenburg et al, 1983; Cooke and Schneider, 1983). Figure 1 is a summary of the currently available seismic inversion methods. The most complete approach to inversion involves the use of pre-stack data, and two forms of pre-stack inversion are recognized: travel-time inversion (often called tomography) and amplitude inversion (a simplification of which is AVO analysis). Post-stack inversion, however, remains the most popular current method, probably because of its greater robustness and simpler assumptions. As seen in Figure 1, there are two major approaches to post-stack inversion, broadband inversion and bandlimited inversion. The broadband method has been further subdivided into the sparse-spike method and the model-based method. These three types of post-stack inversion, bandlimited, sparse-spike and model-based, are discussed in detail by the three papers referred to above, and will be briefly summarized here.

First, we will summarize the basic theory behind all post-stack methods. All are based on the assumption that the seismic trace can be modeled with the convolutional equation, which can be written

$$s_t = [r_t * w_t + n_t] \cdot a_t, \qquad (1)$$

where s_t = the seismic trace,
r_t = the earth's reflectivity,
w_t = the seismic wavelet,
n_t = additive noise,
a_t = amplitude scaling,
$*$ denotes convolution,
and $\cdot$ denotes multiplication.

Equation (1) suggests that if we are able to remove (or reduce) the noise component, deconvolve the wavelet, and restore the original amplitudes (i.e. remove the amplitude scalar), we would be left with the earth's normal incidence reflectivity, which is related to acoustic impedance by the equation

$$Z_{t+1} = Z_t \left[\frac{1 + r_t}{1 - r_t} \right] \qquad (2)$$

where $Z_t = \rho_t V_t$ = acoustic impedance of layer t,
ρ_t = density,
V_t = compressional wave velocity,
and layer t overlies layer t+1.

In practice, we can never exactly recover the reflection coefficients from the seismic trace, and there will always be amplitude, noise and residual wavelet problems. All post-stack inversion methods listed in Figure 1 are therefore approximations of the ideal situation. The general concept of post-stack inversion is shown in the flowchart in Figure 2. The two inputs to inversion are stacked seismic data and a set of geological constraints in the form of a model. These are combined in some way to produce the final inversion. The way in which the information is combined depends on the algorithm used. In straightforward terms, bandlimited inversion involves integrating the seismic data directly to produce a bandlimited inverted trace and then deriving the missing low frequency trend from the geological model. The sparse-spike inversion method involves estimating a set of sparse reflection coefficients from the seismic data, constraining these reflection coefficients with the model and then inverting the coefficients produce the impedance. In model-based inversion, we start with an initial model of the earth's geology and perturb this model until the derived synthetic seismic section best fits the observed seismic data. It should be pointed out that the results of both the sparse-spike and model-based methods are similar, and have the appearance of a "blocky" looking impedance. The differences will be discussed in the next two sections.

MODEL RESULTS:

Two models were used to test the three inversion algorithms. The first was a 6-trace synthetic seismogram derived from the sonic and density logs from a well which encountered a carbonate reef. Since the geological answer is known, the outputs of the various inversion algorithms could be judged based on the known result. Two separate geological constraints were used, one which was close to the known answer, and a second which was a smoothed version of the known answer. As expected, the inversion results degraded when the poorer initial model was used. Specific results for the three inversion methods were as follows. The bandlimited inversion produced a frequency limited version of the sonic log with smoothed interface boundaries. The sparse-spike inversion was sensitive to the detection threshold used for detecting the spikes and therefore missed some of the lower amplitude reflection coefficients. However, the interfaces were much better defined than in the bandlimited inversion. Finally, the model-based inversion produced a similar result to the sparse-spike method, but defined more blocks in the final result.

The second model dataset used was the classic wedge model. In this case, we also used two different geological models as constraints, one which was close to the actual result and the other which was a constant thickness block. It was observed that bandlimited inversion and model-based inversion were both sensitive to the starting model, but sparse-spike inversion was able to overcome the limitation of an incorrect geological constraint. Of the three methods, bandlimited inversion performed the worst and sparse-spike inversion performed the best. Model-based inversion did not perform as well as sparse-spike inversion because of the difficulties in converging to a sparse model. Bandlimited inversion produces poor results because of the noticeable wavelet effects.

REAL DATA RESULTS:

Two real datasets were used to test the inversion algorithms, a gas sand and a carbonate reef. The gas sand example was a classic "bright spot", and was taken from an area where AVO inversion works very well. The objective of doing post-stack inversion was therefore to estimate the lateral changes in thickness and acoustic impedance of the sand. In all cases, the model used was a vertically and laterally stretched version of the acoustic impedance derived from the producing well. As expected, bandlimited inversion produced a smoothed estimate of the sand. The sparse-spike and model-based methods both produced similar looking blocky inversions, but the lateral continuity on the model-based inversion was better.

The second dataset was a carbonate reef in which the lateral extent of a high porosity zone within the reef was the objective. Again, the geological constraint used was a vertically and laterally stretched model based on the acoustic impedance derived from a well log that had been drilled into the reef. The results are shown in Figures 3, 4 and 5 for a few traces close the well. All three methods produced good estimates of the reef porosity. The bandlimited method, shown in Figure 3 produced the most laterally continuous results, whereas the results from the model-based and sparse-spike methods were more discontinuous in a lateral sense. However, the sharpness in the geological interfaces came through much more clearly in both of the "blocky" methods, and a better match to the original log was obtained. The sparse-spike results shown in Figure 4 were less detailed than the model based results. The best overall fit was found using the model-based method, as seen in Figure 5.

CONCLUSION:

Three different post-stack seismic inversion algorithms, were used to invert four datasets, two of which were models and two of which were real examples. It was found that the most robust of the three methods was the classical bandlimited approach. However, this method produced a smoothed, frequency limited estimate of the impedance which was not as sharp in detail as the other two methods. The bandlimited method also failed in the case of a very "sparse" model. The sparse-spike and model-based methods produced visually similar results. The sparse-spike approach produced superior results on a totally "sparse" model, but produced less detailed results than model-based inversion when applied to real data. Although the model-based inversion method is the most intuitively appealing, it has to be carefully constrained to avoid the problem of non-uniqueness.

This study showed that there is no absolutely right way to do post-stack inversion. However, it suggested types of data in which one inversion may be preferred over another.

REFERENCES:

Lindseth, R.O., 1979. Synthetic sonic logs - a process for stratigraphic interpretation: Geophysics, v. 44, p. 3-26

Oldenburg, D.W., Scheuer, T., and Levy, S., 1983. Recovery of the acoustic impedance from reflection seismograms: Geophysics, v. 48, p. 1318-1337

Cooke, D.A., and Schneider, W.A., 1983. Generalized linear inversion of reflection seismic data: Geophysics, v. 48, p. 665-676

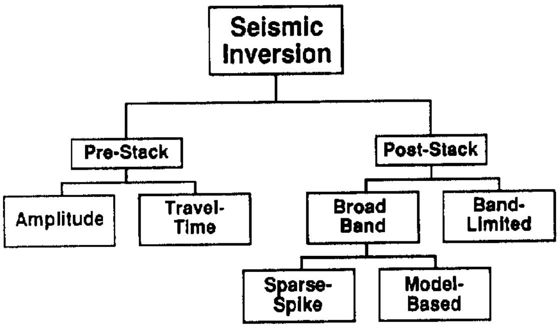

Figure 1 *A summary of current seismic inversion methods.*

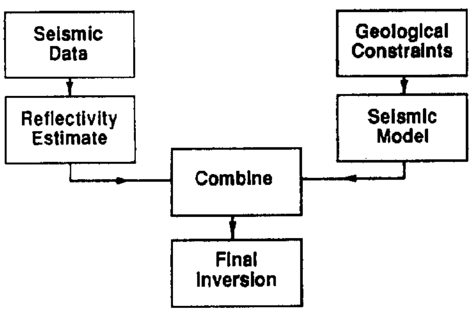

Figure 2 *The general method of post-stack seismic inversion.*

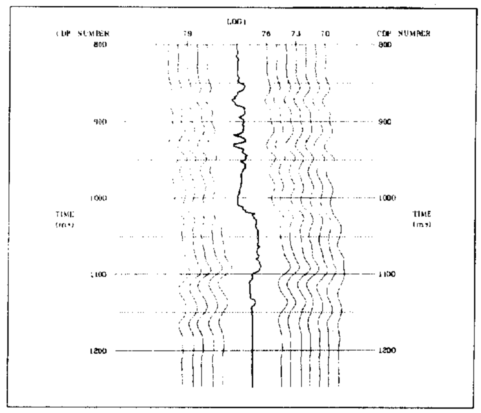

Figure 3 *Bandlimited inversion applied to carbonate reef example. Acoustic impedance from well is shown as heavy trace.*

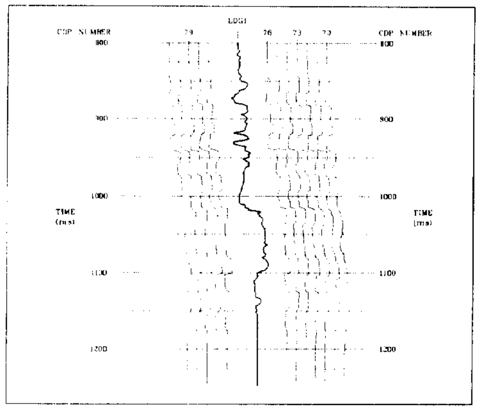

Figure 4 *Sparse-spike inversion of carbonate reef example. Acoustic impedance from well is shown as heavy trace.*

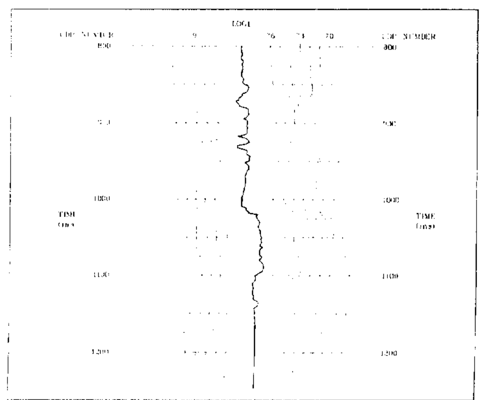

Figure 5 *Model-based inversion of carbonate reef example. Acoustic impedance from well is shown as heavy trace.*